ModelIt!

Teacher's Guide

Where Does Your Electricity Come From?

Tracing Energy from Natural Resources to Your Light Switch

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-ESS3-1

Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment.

Three-Dimensional Learning

Science Practice	**Obtaining, Evaluating, and Communicating Information** — Students gather and combine information about different energy sources and their environmental impacts to make informed decisions.
Disciplinary Core Idea	**ESS3.A Natural Resources** — Energy and fuels that humans use are derived from natural sources, and their use affects the environment in multiple ways. Some resources are renewable over human lifetimes, and some are not.
Crosscutting Concept	**Cause and Effect** — Students identify cause and effect relationships between energy source availability, power production, and electricity delivery to homes.

CAST Testing Alignment

MULTIPLE CHOICE:

A town uses only solar panels for electricity. During a week of cloudy weather, what problem would the town face, and which addition would best solve it?

CONSTRUCTED RESPONSE:

Compare two energy sources — coal and wind — by explaining where each comes from, how it makes electricity, and how it affects the environment.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Electricity is so common in modern life that we rarely think about where it comes from. Yet the journey from natural resource to wall outlet is a fascinating chain of energy conversions with significant environmental implications.

How Power Plants Work

Most power plants work by spinning a turbine connected to a generator. In fossil fuel plants, burning coal or natural gas heats water to create steam, which spins the turbine. In wind farms, moving air spins the turbine directly. In hydroelectric dams, falling water spins the turbine. Solar panels are different — they convert sunlight directly into electricity using photovoltaic cells without any spinning parts.

Fossil Fuels

Coal, natural gas, and oil are fossil fuels formed from ancient plants and animals buried millions of years ago. They store enormous amounts of chemical energy, which is released when burned. However, burning fossil fuels releases carbon dioxide (CO2) and other pollutants into the atmosphere, contributing to climate change and air quality problems. Fossil fuels are nonrenewable — once used, they take millions of years to form again.

Renewable Energy Sources

Solar, wind, hydroelectric, and geothermal energy are renewable — they replenish naturally. Solar power from the sun and wind power from moving air are the fastest-growing energy sources worldwide. Their main limitation is intermittency: solar panels do not produce electricity at night, and wind turbines need wind to spin. Battery storage technology is rapidly improving to address this challenge.

The Power Grid

Electricity travels from power plants to homes through the power grid — a vast network of transmission lines, transformers, and substations. High-voltage transmission lines carry electricity over long distances, and transformers step down the voltage for safe home use. The grid must constantly balance supply and demand because electricity cannot be easily stored in large quantities, though battery technology is changing this.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify energy source amount, power plant output, and home electricity as the key components of the electricity generation and delivery system.

E **Establish Relationships**
Students determine that energy source amount positively affects power plant output (more fuel or sunlight = more electricity produced), and power plant output positively affects home electricity (more production = more electricity available for homes).

V **Visualize & Model**
Students build a model showing how natural resources flow through power plants to become the electricity that powers homes.

E **Evaluate Outcomes**
Students run sunny day, cloudy week, and windy night scenarios to observe how energy source availability affects the entire electricity chain.

R **Revise & Extend**
Students add battery storage or transmission distance to explore more realistic electricity system dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, electricity is flowing through the walls around you. But where did it come from?"*

Watch For

Turn off the classroom lights for 3 seconds, then turn them back on. Ask: What just happened?

Slide 2: Learning Objectives

2 min

Say: *"Today we are going to trace electricity from its source all the way to your light switch."*

Watch For

Have students read objectives aloud. Pre-teach natural resource and renewable energy.

Slide 3: Big Question

3 min

Say: *"Think about everything you used electricity for today. Now — where did ALL that electricity come from?"*

Watch For

Have students list 5 things they used electricity for. Then ask: But where does it START?

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to understand the journey of electricity from nature to your home."*

Watch For

Preview LEVER steps. Build curiosity about the energy chain.

Slide 5: Activity 1: Components

7 min

Say: *"Every light switch connects back to a natural resource. Let us figure out the chain."*

Watch For

Guide component sorting. Discuss why energy sources come from outside the system.

Slide 6: Activity 2: Connections

7 min

Say: *"More sunlight means more electricity from solar panels. More electricity from the plant means more power for homes. Both positive!"*

Watch For

Help students draw relationship arrows. Both relationships are positive in this system.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens to your electricity when a cloud covers the sun? Let us find out!"*

Watch For

Students predict, then run scenarios. Compare sunny day vs cloudy week.

Slide 8: Discoveries

4 min

Say: *"Now you know — every time you flip a switch, nature is working to power it!"*

Watch For

Lead discussion about trade-offs between energy sources. Show real-world examples.

Slide 9: STEM Challenge

8 min

Say: *"Your town needs a new energy plan. You decide how to power 1,000 homes!"*

Watch For

Distribute materials. Groups design energy plans balancing reliability and environment.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Energy Source Amount
- **INTERNAL:** Power Plant Output, Home Electricity

Relationships

- **Energy Source Amount to Power Plant Output:** POSITIVE (+) — More available energy source (more sunlight, more wind, more coal) means the power plant can produce more electricity. Less source means less output.
- **Power Plant Output to Home Electricity:** POSITIVE (+) — When the power plant produces more electricity, more is available to flow through power lines to homes. When output drops, homes may experience brownouts or blackouts.

Key Discoveries

- All electricity comes from converting natural resources — we cannot create energy from nothing
- More available energy source means more power plant output and more home electricity
- Renewable sources like solar and wind depend on weather, while fossil fuels can be burned anytime
- Every energy source affects the environment — fossil fuels create pollution, while renewables need space for panels and turbines

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Electricity comes from the wall outlet"	The outlet is just the final delivery point. Electricity is generated far away at a power plant by converting natural resources into electrical energy. It then travels through miles of power lines and multiple transformers to reach your home. The outlet is like a faucet — the water does not come from the faucet, it comes from a water source far away.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Renewable energy is free"	While sunlight and wind are free, the equipment to capture them is not. Solar panels, wind turbines, and batteries all cost money to build, install, and maintain. However, once installed, the ongoing cost is much lower than fossil fuels because you do not need to keep buying fuel. Over time, renewable energy often becomes cheaper than fossil fuel energy.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can just turn off power plants when we do not need electricity"	Power plants cannot turn on and off like a light switch. Coal and nuclear plants take hours or even days to start up and shut down. This is why the power grid must constantly balance supply and demand in real time. Solar and wind add complexity because their output varies with weather. Grid operators work 24/7 to keep everything balanced.